Assignment 5 - SQL databases - C++

3/3/2022

Legg til kommentar 0

5/6/2022

Detaljer

Introduction

- Before you start make sure you have completed Course preparations.
- Each assignment below is a separate directory. When you create each project the directory path will be *ikt103g22v/assignments/solutions/assignment_?_?*.

5.1 Using a database

Create a program that displays the following menu to the user:

1. Read all students
2. Get student by id
3. Add student
4. Edit student
5. Remove student
6. Search
7. Exit

then performs the choice entered by the user. You can reuse menu code from the previous assignment if you want to.

All data must be stored in the SQLite database called *school.sqlite*.

The *students* table has the following columns:

- **id** integer primary key auto increment
- **name** text
- **email** text
- **year** integer

Areas of interest: SQLitecpp, for loops

1. Read all students

Print information about every student stored in the database.

Expected output:

- *id: 1, name: Kathryn Boyd, email: kathrynb@uia.no, year: 2018*

  ...
- *No students found*

2. Get student by id

Asks the user for a student id then retrieves that student from the database.

Expected output:

- *id: 1, name: Kathryn Boyd, email: kathrynb@uia.no, year: 2018*
- *Student not found*

3. Add student

Asks the user for the following (in order):

1. Name
2. Email
3. Year

Then adds the new student to the database.

Expected output: *Added student: id: 1, name: Kathryn Boyd, email: kathrynb@uia.no, year: 2018*

4. Edit student

Asks the user for the following (in order):

1. Student id
2. Name
3. Email
4. Year

Then updates the existing student in the database.

Expected output:

- *Student was edited successfully*
- *Student not found*

5. Remove student

Asks the user for a student id then removes that student from the database.

Expected output:

- *Student was removed successfully*
- *Student not found*

6. Search

Asks the user for a string then searches for students with matching names. The user can use (or not use) % themselves, so don't add it in code.

Example user inputs:

- Lise Larsen
- Knut%
- %Andersen
- %arn%

Expected output:

Same as in choice 1.

7. Exit

Exits the program.

Testing and delivery

see *Local testing and Delivery and Bamboo testing* under Assignments.